

RECOMMENDER SYSTEMS & SECURITY

Guarding the Recommendation Pipeline

SVD/ALS Factorization with Fake Review Filtering

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Domain: Matrix Factorization & NLP Security

Problem & Core Challenges

▣ Extreme Data Sparsity

E-commerce rating matrices $R \in \mathbb{R}^{m \times n}$ are inherently sparse because consumers interact with only a tiny fraction of global inventories.

Standard recommendation algorithms struggle to uncover latent user preferences or predict missing ratings accurately under high sparsity.

🛡️ Deceptive Feedback Attacks

AI-generated fake reviews, bot activities, and rating inflation manipulate missing rating predictions before factorization occurs.

Unfiltered malicious data severely distorts matrix decomposition, leading to biased and untrustworthy product recommendations.

Matrix Factorization Engine



Truncated SVD

Decomposes sparse matrices into low-rank representations to map users and items into a shared latent factor space, capturing complex hidden relationships.



Regularized ALS

Optimizes Alternating Least Squares with regularized penalty parameters to minimize iterative approximation error and prevent model overfitting.



Item-Centric Query

Takes specific product inputs to extract a curated list of top-rated, highly relevant alternatives based on robust latent similarity computations.

Fake Review NLP Guardrail

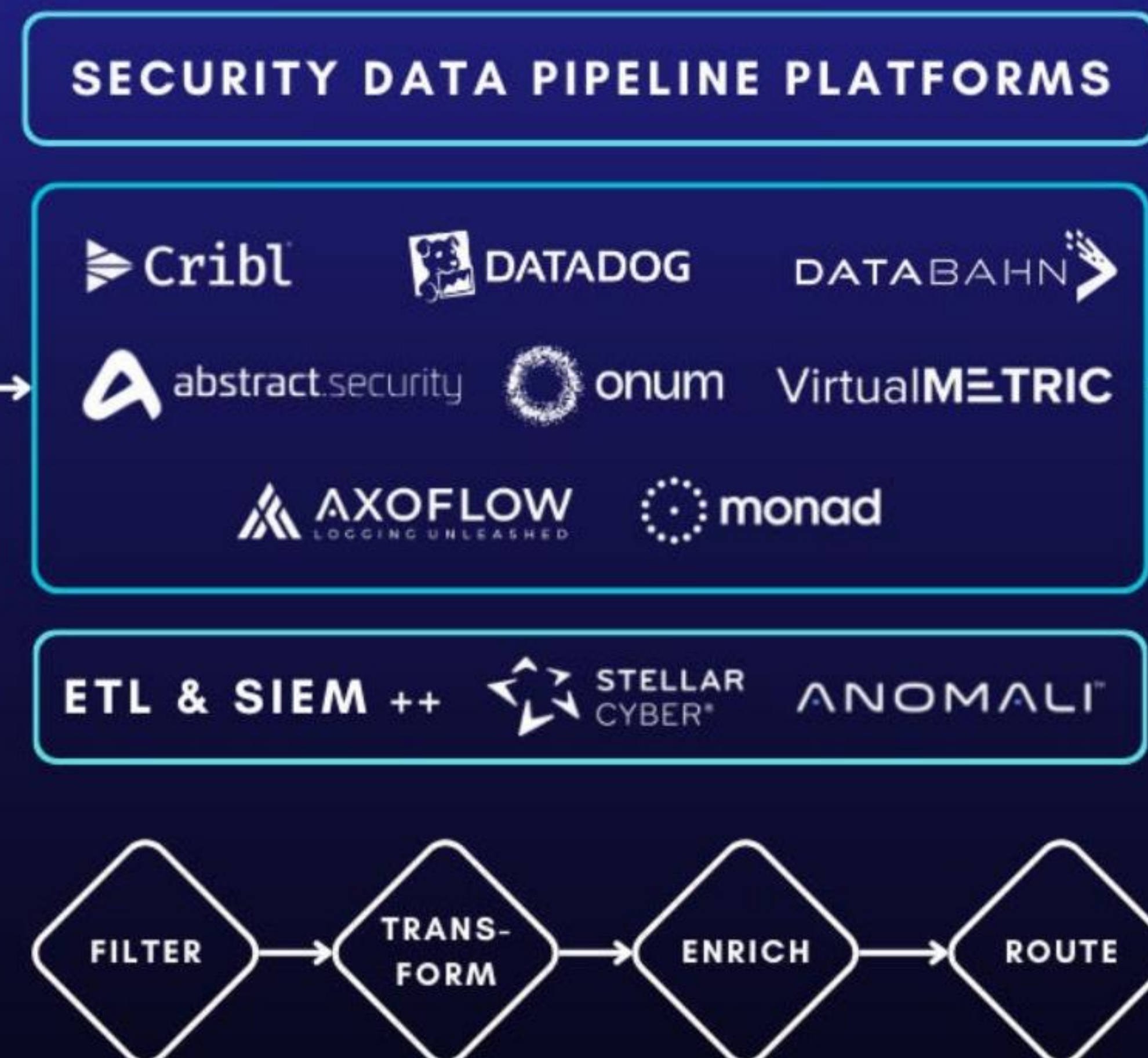
Input-Stage Security Layer

An integrated NLP and Machine Learning guardrail acts as a proactive security layer before collaborative filtering processes the preference matrix.

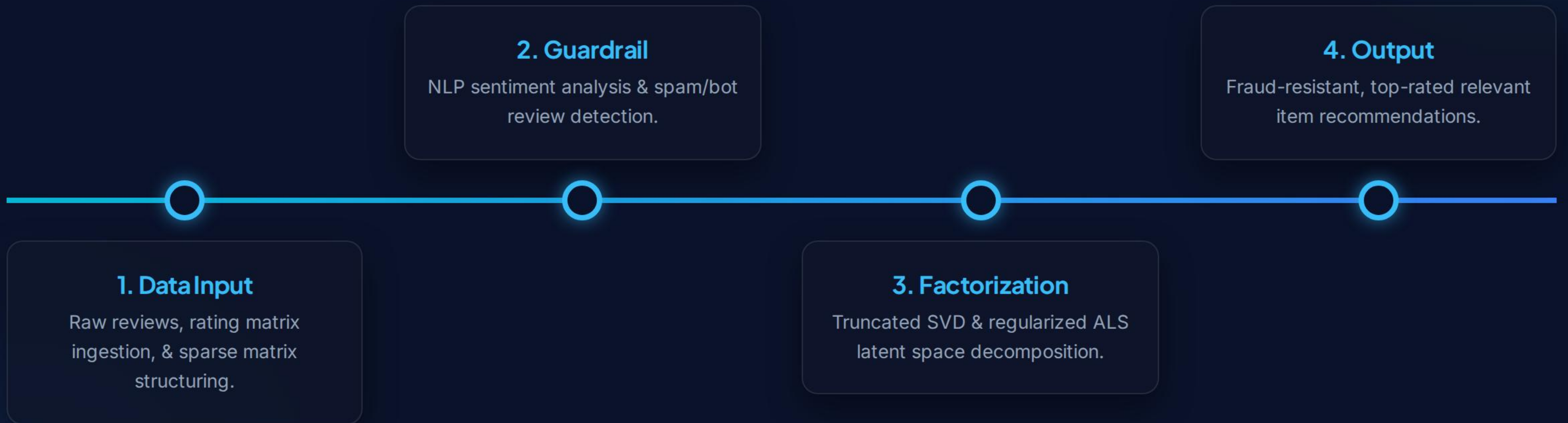
The guardrail detects and neutralizes bot-like footprints, fraudulent feedback, and sentiment-rating anomalies in real-time.

By filtering out rating inflation early, the factorization engine reflects true, organic user preference patterns.

LAYER OF THE MODERN SOC PL



System Pipeline Workflow



Performance & Evaluation

RMSE
& MAE Error Metrics

Rigorously Evaluated Architecture

Evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) across varying latent dimensions and regularization penalties.

Optimized Python implementation utilizing scipy sparse matrices for high scalability.

Successfully bridges classical numerical linear algebra with supervised learning to deliver a robust, fraud-resistant engine for large marketplaces.

Image Sources



[https://substackcdn.com/image/fetch/\\$s_!pSR2!,f_auto,q_auto:good,fl_progressive:steep/https%3A%2F%2Fsubstack-post-media.s3.amazonaws.com%2Fpublic%2Fimages%2F5244cbee-8937-468d-b1b6-758cbcafbcb75_1920x1080.png](https://substackcdn.com/image/fetch/$s_!pSR2!,f_auto,q_auto:good,fl_progressive:steep/https%3A%2F%2Fsubstack-post-media.s3.amazonaws.com%2Fpublic%2Fimages%2F5244cbee-8937-468d-b1b6-758cbcafbcb75_1920x1080.png)

Source: softwareanalyst.substack.com